How Sparkling Wine Is Made

How the bubbles get into the bottle — the four French roads to sparkling wine. Tap any step to open the details, hover the **underlined terms** for a quick definition.

Champagne

1. Harvest *Vendange*

Champagne is **hand-harvested** by law, in whole bunches, from Chardonnay, Pinot Noir and Meunier. Grapes are picked **early**, while acidity is high and sugar modest — exactly what a sparkling base wine needs.

Why it matters: high acidity and low alcohol are the backbone of a wine that must survive a second fermentation and long ageing.

2. Whole-bunch Pressing *Pressurage*

Whole bunches are pressed gently in a wide **coquard** press. Even the black Pinots give nearly colourless juice; yields are strictly capped, and only the finest early fraction (the *cuvée*) goes into the top wines.

Why it matters: gentle pressing keeps the juice white and free of harsh tannin — essential for an elegant sparkling wine.

3. First Fermentation *Fermentation alcoolique*

The juice ferments into a dry, sharp, still **base wine** (*vin clair*) of only 10–11% alcohol. Each grape, village and plot is usually kept separate for now.

Why it matters: this austere still wine is the blank canvas the whole method is built on.

4. Assemblage *Assemblage*

The cellar master blends dozens of base wines — grapes, villages, and often older **reserve wines** — into the final **assemblage**. This is where a house's signature is created, and how non-vintage Champagne tastes consistent every year.

Why it matters: blending, not a single vineyard, is the art at the heart of Champagne.

5. Tirage *Mise en tirage*

The blend is bottled with a **liqueur de tirage** (wine, sugar and yeast) and sealed under a crown cap. This is the **tirage** — the trigger for the magic to come.

Why it matters: from here the bubbles are made *inside this very bottle* — the defining idea of the method.

6. Second Fermentation in Bottle *Prise de mousse*

Laid on their sides, the bottles undergo a slow second fermentation — the **prise de mousse**. The CO_2 cannot escape the sealed bottle, so it dissolves into the wine as fine bubbles, building around 6 bar of pressure.

Why it matters: this in-bottle fermentation is what separates **méthode traditionnelle** from every quicker method.

7. Ageing on the Lees *Élevage sur lie*

The wine then rests **sur lie** — on the dead yeast **lees**. Slow **autolysis** gives Champagne its toast, brioche and biscuit. The legal minimum is 15 months (3 years for vintage); great houses age far longer.

Why it matters: lees ageing is where fizzy wine becomes *Champagne* — complex, creamy and deep.

8. Riddling *Remuage*

Each bottle is slowly tilted neck-down and twisted — **remuage** — until all the **lees** collect against the cap. Once done by hand on wooden *pupitres*, now usually by a **gyropalette**.

Why it matters: gathering the lees in the neck makes it possible to remove them cleanly next.

9. Disgorgement *Dégorgement*

The neck is dipped in freezing brine to trap the lees in a little ice plug; the cap is removed and the bottle's own pressure fires the frozen pellet out — **dégorgement** — leaving perfectly clear wine.

Why it matters: it removes the sediment without losing the precious bubbles.

10. Dosage & Corking *Dosage*

The tiny gap is topped up with the **liqueur d'expédition** — the **dosage** — which fixes the style from bone-dry **brut** Nature to sweet Doux. Then the mushroom cork, wire **muselet** and foil go on.

Why it matters: a few grams of sugar here decide whether the wine tastes racy or rounded.

11. Ageing & Release *Repos & expédition*

The finished bottles rest a few more months for the dosage to marry in, then they are labelled and released — ready to open with a soft ‘sigh’ (never a violent pop, which wastes the mousse).

Why it matters: a short final rest lets everything integrate so the first sip sings.

Crémant

1. Harvest *Vendange*

Crémant is made by the **méthode traditionnelle** **outside** Champagne — in Alsace, the Loire, Burgundy, Limoux, Bordeaux, Die, Jura and Savoie. Like Champagne it must be **hand-harvested in whole bunches**, from each region's own grapes (Chenin, Chardonnay, Pinot, Riesling, Mauzac…).

Why it matters: the hand-harvest and whole-bunch rule lets Crémant claim the same quality method as Champagne.

2. Whole-bunch Pressing *Pressurage*

As in Champagne, whole bunches are pressed gently, with limited juice yields, to give clean, pale base juice.

Why it matters: the same gentle pressing keeps Crémant fresh and fine rather than coarse.

3. First Fermentation *Fermentation alcoolique*

The juice ferments into a dry, high-acid **base wine**, just as for Champagne.

Why it matters: a crisp base wine is the foundation for lively bubbles.

4. Assemblage *Assemblage*

Base wines are blended into the cuvée — often expressing one region's grapes rather than the many-village blends of Champagne.

Why it matters: blending balances the wine and sets each producer's style.

5. Tirage *Mise en tirage*

Bottled with the **liqueur de tirage** and crown-capped to start the second fermentation — identical to Champagne's **tirage**.

Why it matters: the in-bottle fermentation is exactly what makes it a ‘traditional method’ wine.

6. Second Fermentation in Bottle *Prise de mousse*

The **prise de mousse** takes place in each sealed bottle, trapping the CO_2 as fine bubbles.

Why it matters: same physics as Champagne — real, fine, long-lasting mousse.

7. Ageing on the Lees *Élevage sur lie*

Crémant must rest **sur lie** for at least **9 months** (versus 15 for Champagne) — one of the few real differences. Many producers age well beyond that for extra **autolysis** character.

Why it matters: the shorter minimum is part of why good Crémant offers the Champagne method at a friendlier price.

8. Riddling *Remuage*

Bottles are riddled — **remuage** — by hand or **gyropalette** to gather the lees in the neck.

Why it matters: the same clarity-preparation step as Champagne.

9. Disgorgement *Dégorgement*

The frozen lees plug is disgorged — **dégorgement** — under pressure, leaving clear wine.

Why it matters: removes sediment while keeping the sparkle.

10. Dosage & Corking *Dosage*

A **dosage** sets the style, then the cork, **muselet** and foil are fitted.

Why it matters: the finishing touch that balances the wine.

11. Release *Expédition*

After a short rest the Crémant is labelled and released — often superb value for a genuine bottle-fermented sparkler.

Why it matters: proof that the Champagne method travels well across France.

Méthode Ancestrale

1. Harvest *Vendange*

The oldest sparkling method of all — older than Champagne. Grapes such as Mauzac (Limoux), Gamay (Bugey-Cerdon) or Muscat are picked with enough natural sugar to carry a single, unfinished fermentation.

Why it matters: everything depends on the grape's own sugar — nothing is added later.

2. Pressing *Pressurage*

The grapes are pressed and the juice run to tank to begin a single fermentation.

Why it matters: a clean start to the one and only fermentation this wine will get.

3. Single Fermentation begins *Fermentation unique*

Fermentation starts in tank, often cool and slow. Crucially it is **not allowed to finish** — the winemaker watches the sugar fall to just the right point.

Why it matters: timing is everything: too soon and the wine is sweet and gassy, too late and it is flat.

4. Early Bottling *Mise en bouteille*

While it is still fermenting, the wine is bottled and sealed — usually under a crown cap. **No yeast and no sugar are added**; the remaining natural sugar is all it needs.

Why it matters: this single, uninterrupted fermentation is what defines **méthode ancestrale**.

5. Fermentation Finishes in Bottle *Prise de mousse*

The last of the fermentation happens in the sealed bottle, trapping CO_2 as a soft, natural sparkle. Pressure is usually lower than Champagne's, and the wine is often cloudy and gently sweet.

Why it matters: the charm of a **pétillant naturel** is exactly this rustic, lightly-fizzy freshness.

6. Disgorge or Leave Cloudy OPTIONAL *Dégorgement (facultatif)*

Some producers disgorge the lees for a clearer wine; many leave them in for a hazy, characterful 'pét-nat'. Either way there is **no dosage** — the wine is bottled as nature left it.

Why it matters: it is a stylistic choice, not a rule — clarity versus rustic authenticity.

7. Release *Expédition*

Ancestral-method wines — Limoux's Blanquette ancestrale, Bugey-Cerdon, Gaillac, Clairette de Die — are best enjoyed young, lightly chilled.

Why it matters: freshness and immediacy, not long ageing, are the point.

Méthode Charmat

1. Harvest *Vendange*

The tank method suits **aromatic** grapes whose fresh, floral scent you want to keep — Glera (for Prosecco), Muscat (for Asti) and many fruity styles.

Why it matters: the whole method is designed to preserve primary fruit and flower aromas.

2. Base Wine *Vin de base*

The grapes are pressed and fermented into a fresh, fruity still **base wine**, as usual.

Why it matters: a clean, aromatic base is what the tank method aims to keep intact.

3. Second Fermentation in Tank *Cuve close*

Here is the big difference: the second fermentation happens in a large sealed, pressurised steel tank (*autoclave*) — the **méthode charmat** (or *cuve close*). The whole batch bubbles together, in weeks rather than years.

Why it matters: fast, cool tank fermentation locks in fresh grapey aromas that long lees-ageing would mask.

4. Brief Lees Contact OPTIONAL *Sur lie (court)*

Most Charmat wines get little or no **lees** ageing, to stay bright and fruity — though a 'Charmat lungo' style rests longer for a touch more complexity.

Why it matters: keeping lees contact short is deliberate — it preserves the fresh, primary character.

5. Filter under Pressure *Filtration isobare*

The sparkling wine is chilled and filtered **under pressure** so it stays clear and keeps its fizz.

Why it matters: filtering under pressure removes the yeast without letting the bubbles escape.

6. Bottling & Dosage *Tirage isobare*

Using a counter-pressure filler, the wine is bottled under pressure and given its **dosage** to set the sweetness (Prosecco is often Extra Dry; Asti sweeter).

Why it matters: isobaric bottling is the trick that gets the fizz safely from tank to bottle.

7. Release *Expédition*

Charmat sparklers are made to be drunk young, when their fresh fruit and flowers are at their most vivid.

Why it matters: these wines are about immediate, joyful freshness — not cellar age.

The four roads to bubbles

Champagne — Méthode Traditionnelle

Champagne is the benchmark for the **méthode traditionnelle**: a still **base wine** is blended (**assemblage**), bottled with a **liqueur de tirage**, and fermented a **second time inside the bottle** (**prise de mousse**). Long ageing **sur lie**, then **remuage**, **dégorgement** and **dosage** finish the wine. Only grapes grown and hand-harvested in the Champagne region may carry the name.

Crémant — the method beyond Champagne

Crémant is the very same method used **outside** Champagne — Alsace, Loire, Bourgogne, Limoux, Bordeaux, Die, Jura, Savoie. Hand-harvest and whole-bunch pressing are required; the main difference is a shorter minimum ageing **sur lie** (9 months). Often excellent value for a true bottle-fermented sparkler.

Méthode Ancestrale — the original ‘pét-nat’

The **méthode ancestrale** is older than Champagne and far simpler: a **single** fermentation is bottled before it finishes, trapping natural CO_2 with no added yeast or sugar and no **dosage**. Today's **pétillant naturel** revives it — lightly fizzy, often cloudy. Classics: Blanquette de Limoux ancestrale, Bugey-Cerdon, Clairette de Die, Gaillac.

Méthode Charmat — the tank method

The **méthode charmat** (tank method / *cuve close*) runs the second fermentation in a big **pressurised tank** instead of the bottle, then bottles under pressure. Fast and fruit-forward, it is ideal for aromatic grapes — the method behind Prosecco and Asti.

Glossary

Assemblage — blending
Blending different grapes, villages and vintages — often with older reserve wines — to build a consistent house style before bottling.

Autolysis
The slow breakdown of dead yeast cells during lees ageing, releasing the toasty, brioche flavours prized in traditional-method sparkling wine.

Base wine — *vin de base / vin clair*
A dry, tart, low-alcohol still wine — the raw material for every sparkling wine before the bubbles are added.

Brut & the sweetness scale
Dosage sets the style: **Brut Nature** (0–3 g/L, nothing added), **Extra Brut**, **Brut** (<12 g/L), **Extra Dry**, **Sec**, **Demi-Sec**, **Doux** (sweetest).

Coquard press
The wide, shallow traditional Champagne press that gently squeezes whole bunches, keeping the juice pale even from black grapes.

Dégorgement — disgorgement
Removing the plug of lees: the neck is frozen, the cap flicked off, and the frozen pellet shoots out under the bottle's own pressure.

Dosage
The final touch of wine and sugar added after disgorgement, which sets the wine's sweetness.

Gyropalette
A computer-controlled cage that riddles hundreds of bottles at once — days of work where hand-remuage took weeks.

Lees — *lie*
The spent yeast cells left after fermentation. Long contact with them (autolysis) adds creamy, bready complexity.

Liqueur d'expédition
The wine-and-sugar mixture used for dosage; how much sugar it carries decides whether the wine is Brut, Extra Brut, and so on.

Liqueur de tirage
A dose of wine, sugar and yeast added at bottling — the fuel for the second fermentation that makes the bubbles.

Méthode Ancestrale
The oldest way to make sparkling wine: a **single** fermentation, bottled before it finishes so the last of the CO_2 is trapped. No added yeast or sugar.

Méthode Charmat — *cuve close*
The tank method: the second fermentation happens in a large **pressurised tank**, not the bottle. Fast and fruit-forward — how Prosecco and Asti are made.

Méthode Traditionnelle — *Champenoise*
The classic method where the bubbles come from a **second fermentation inside the bottle**. Used for Champagne, Crémant, Cava and more.

Muselet
The wire cage holding the mushroom cork against the bottle's pressure — about 6 bar, three times a car tyre.

Pétillant Naturel — 'pét-nat'
The modern name for méthode ancestrale wines: lightly fizzy, often cloudy and off-dry, sealed under a crown cap.

Prise de mousse
'Capturing the sparkle' — the second fermentation in the sealed bottle, whose CO_2 cannot escape and dissolves into the wine as fine bubbles.

Remuage — riddling
Gradually tilting and twisting each bottle neck-down so the lees slide into the neck. Done by hand on *pupitres* or by machine (gyropalette).

Reserve wines — *vins de réserve*
Still wines held back from earlier harvests and blended in to keep a non-vintage cuvée tasting the same year after year.

Sur lie — on the lees
Ageing the wine in contact with the dead yeast cells; the source of Champagne's toast, brioche and biscuit character.

Tirage
Bottling the blended base wine with the liqueur de tirage and a crown cap, to trigger the second fermentation.

Interactive guide to French sparkling wine · A Quick Glance at French Wine · Chris Paul.
Works fully offline — a single self-contained file.